# Measuring and Validating Risk on a Multi-Asset Portfolio

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github.com/SebastianIWright/multi-asset-risk-dashboard

## The Question

A risk desk needs two answers every day: how much can this portfolio lose, and can we trust that number? I built a daily risk engine for a $1M portfolio of equities, bonds, gold, and an options overlay, then tested whether its VaR model actually held up.

## Approach

- Computed 1-day VaR four ways: historical simulation, parametric (delta-normal), Monte Carlo with Cholesky-correlated paths, and GARCH(1,1).
- Added CVaR (expected shortfall), plus marginal and component VaR to see which positions drive risk.
- Priced the covered call and protective put with Black-Scholes and computed full Greeks.
- Ran six historical stress scenarios (2008 GFC, COVID, 2022 rate shock, dot-com, flash crash, Euro crisis) and a Kupiec POF backtest with the Basel traffic-light system.

## Results

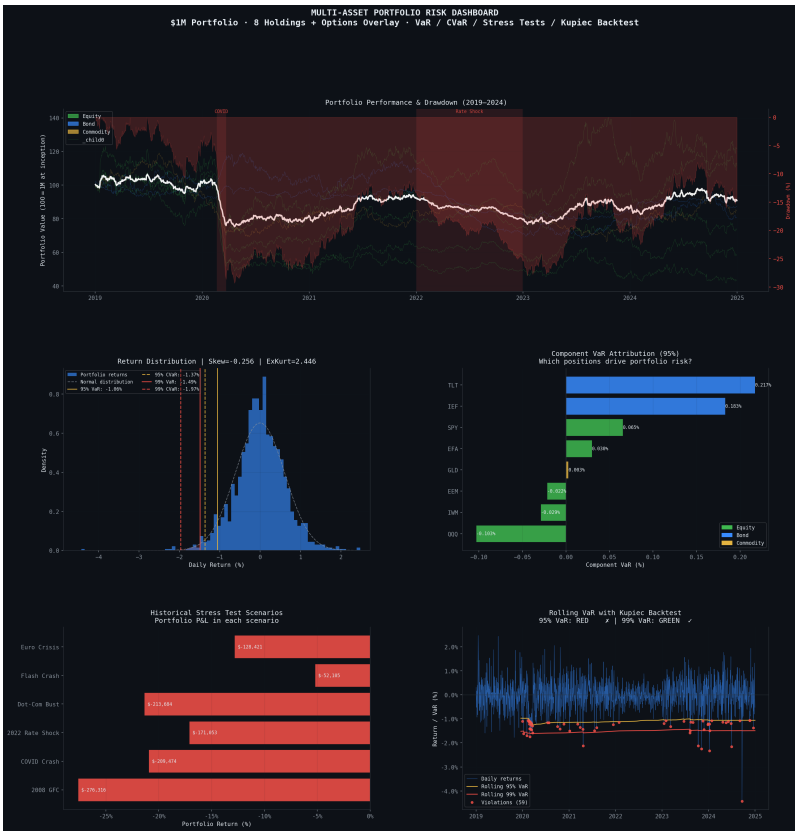
| Metric | Value |
|---|---|
| 1-day 95% VaR (historical) | $10,603 |
| 1-day 99% VaR (historical) | $14,911 |
| 1-day 99% CVaR | $19,704 |
| Annualized volatility | 9.71% |
| Worst stress scenario (2008 GFC) | -27.63% |
| Kupiec backtest, 95% VaR | Red: rejected |
| Kupiec backtest, 99% VaR | Green: accepted |

## Takeaways

- The 95% model failed its own backtest while the 99% model passed. Historical simulation understated risk at 95% for this portfolio, which is exactly what a backtest exists to catch before a model sets real limits.
- Bonds (TLT and IEF) were the largest risk contributors, not equities. Long duration matters even in a 60% equity portfolio.
- Showing the spread across VaR methods is more honest than reporting one number as the truth.

## Next Steps

- Swap the calibrated synthetic return engine for live market data.
- Add 10-day VaR scaling per Basel and the Christoffersen test for clustered violations.




Six-panel daily risk dashboard.